

Mark schemes

Q1.

(a) $\frac{dy}{dx} = -x^2 \sin 2x + \cos 2x$

product rule $kx \sin 2x \pm \cos 2x$

M1

no further incorrect working

A1

2

(b) (i) $-2\alpha \sin 2\alpha + \cos 2\alpha = 0$

replacing $x = \alpha$ and writing equation equal to zero (at any line)

M1

$$\left. \begin{array}{l} 2\alpha \sin 2\alpha = \cos 2\alpha \\ 2\alpha \tan 2\alpha = 1 \\ 2\alpha \tan 2\alpha - 1 = 0 \end{array} \right\} \text{either}$$

AG; CSO

A1

2

(ii) $\left. \begin{array}{l} f(0.4) = 0.2 \\ f(0.5) = -0.6 \end{array} \right\} \text{awrt o.e.}$
(0.9's unsubstantiated scores M0)

M1

Change of sign $\therefore 0.4 < \alpha < 0.5$

A1

2

(iii) $2x \tan 2x = 1$

$$\left. \begin{array}{l} \tan 2x = \frac{1}{2x} \\ 2x = \tan^{-1}\left(\frac{1}{2x}\right) \end{array} \right\} \text{either}$$

$$x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$$

AG; CSO

B1

1

(iv) $x_1 = 0.4$
 $x_2 = 0.4480\dots$
 $x_2 = 25.7$

M1

$x_3 = 0.4200\dots$
 $= 0.42$

A1

2

(c) $y = x \cos 2x$

$$\left. \begin{array}{l} u = x \quad du = 1 \\ dv = \cos 2x \quad v = \frac{\sin 2x}{2} \end{array} \right\}$$

differentiate one term
integrate one term } *must be $k \sin 2x$*

M1

*correct substitution of their values into
 parts formula using $u = x$*

m1

$$\begin{aligned} \int &= \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{4} (dx) \\ &= \left[\frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \right]_{(0)}^{(0.5)} \end{aligned}$$

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A1

$$= \left(\frac{\sin 1}{4} + \frac{\cos 1}{4} \right) - \left(\frac{\cos 0}{4} \right)$$

*correctly substituting values from
 previous 2 method marks*

m1

$= 0.0954$

AWRT

A1

5

[14]